

AI Discovery Questionnaire — Session Script

Use this in the requirements session. Ask each question, offer the cases as "is it like this, this, or this?", write the answer on the > **Answer:** line.

Order: Part A (Purpose) first — it defines what "good" means. Then B (Requirements), then C (Enhancements).

Tags: 🎯 Purpose · 📄 Requirement · ⚡ Enhancement · ⚙️ Infra/On-prem **Full rationale + current build status for every question:** see [AI_REQUIREMENTS_DISCOVERY.md](#).

Part A — Purpose (ask first; defines what "good" means)

A1 — If the AI could do only ONE thing well, what is it? 🎯

- Cut chiller energy cost
- Catch faults/degradation early (before failure)
- Speed up root-cause investigation
- Speed up shift handover / status briefing
- Be a credibility/demo asset to win the next facility

Answer:

A2 — What are the top 5 questions operators actually ask about the plant today? 🎯 *(Capture verbatim — these become the agent modes, the tools we build, and the eval cases.)*

- e.g. "why is chiller 2 using more power?"
- e.g. "which chiller should run as lead?"
- e.g. "is anything about to fail?"
- e.g. "what did the night shift miss?"
- e.g. "are we within efficiency target this week?"

Answer:

A3 — What's the worst outcome if the AI is confidently WRONG? 🎯

- Operator acts on a bad recommendation (needless service call / wrong setpoint)
- Trust collapses → nobody uses it
- A real fault is missed because the AI said "all fine"

Answer:

A4 — How should the AI sound, and in what language(s)? 🎯

- Terse one-liners (floor operators) vs explanatory paragraphs (engineers)
- English-only (enforced today) — final, or multilingual later?

Answer:

A5 — Advisory only, or should the AI eventually ACT? 🎯 ★ *(the defining decision)*

- **A — Advisory only:** reads data, writes advice, no writes anywhere (*current*)
- **B — Soft actions w/ approval:** drafts a work order / ticket / Slack alert → human approves before it's created
- **C — Operator-applied:** AI suggests a setpoint; operator applies it manually in the BMS
- **D — Closed-loop:** AI writes setpoints directly (*considered out of scope/unsafe — confirm it stays out*)

Answer (A/B/C/D + if B+, which actions & who approves):

Part B — Requirements (the must-haves the build must satisfy)

B1 — How non-technical is the LEAST technical user?

- Types clean English → light NL handling
- Types typos/shorthand ("chillor 1 last nite") → need fuzzy matching
- Barely types → need presets/buttons, not free text

Answer:

B2 — Should the agent reason freely or follow guided playbooks?

- **Free ReAct** (*current*): LLM picks which tools to call, in any order — flexible, occasionally wanders
- **Guided playbook:** "if anomaly found → root_cause path; else → brief path" — predictable, auditable, rigid

Answer:

B3 — Is "every claim cited + numbers provably correct" a HARD gate or best-effort?

- Hard gate → build LLM-as-judge + numeric reference check
- Best-effort → current postchecks are enough for now

Answer:

B4 — How much of the AI's reasoning should operators SEE?

- Full live tool trace + audit flags (engineers)
- Just the answer, trace hidden by default (floor operators)
- Confidence level stated on uncertain answers

Answer:

B5 — Is open-source-only firm, with no exceptions ever?

- Firm — data never leaves the building (*current mandate*)
- Open to Azure OpenAI **Private Endpoint** if data stays in our tenant

Answer:

B6 — What GPU and how many concurrent operators?

- 1 box (RTX 4080/4090-class), 1–3 users → Ollama is fine (*current*)
- 10+ users across shifts → vLLM migration justified

- Server GPU budget (A100) → bigger models + fine-tuning possible

Answer:

**B7 — What's the pass bar before a build is "trustable"? **

- Green eval + a demo (*current POC bar*)
- A domain expert reviews N real answers and signs off
- Numeric answers must be provably correct

Answer:

**B8 — What answer latency is acceptable, and what's the degraded-mode expectation? **

- Streaming makes ~30s fine (*current bet*)
- Hard <10s needed → reduce steps / right-size models / vLLM
- On outage: clear "model unavailable" message acceptable vs must alert someone

Answer:

**B9 — What malicious / adversarial inputs must the AI resist? (GenAI security) **

- Prompt injection ("ignore your rules and ...") — guardrails exist, how hard a requirement?
- Jailbreak attempts to get unsafe/off-topic output
- Malformed NL→SQL trying to read other tables
- Realistically a trusted internal LAN tool → low threat, OR exposed to many users → harden

Answer:

**B10 — What must the AI REFUSE or defer to a human? **

- Anything it can't ground in data (refuse vs guess) (*premise-verification exists*)
- Safety-critical procedures / regulatory advice → defer to human expert
- Out-of-domain questions (not about the plant)
- Any action beyond the autonomy ceiling set in A5

Answer:

**B11 — What data can the AI see, how fresh, and is any of it sensitive? **

- Allowed sources: telemetry only, or also tariffs / maintenance logs / production schedule?
- Freshness: real-time (1-min) vs batched (15-min → answers must say "as of HH:MM")
- Sensitive/PII data that must never enter a prompt? (people, contracts, costs)
- Known bad data to defend against (stuck sensors, unit mismatches, clock skew)

Answer:

**B12 — What's the equipment & domain scope, and are the benchmarks agreed? **

- Just the central chiller plant (2 chillers + tower + pump) (*current*)
- Add AHUs / FCUs / pumps / VFDs / boilers → new knowledge + tools
- Whole-facility energy (compressed air, lighting)
- Are the kW/TR design values & efficiency bands per equipment confirmed correct by the facility?

Answer:

B13 — Who can change prompts/models, and how do we roll back? (AI governance) 📌

- Prompt change control: anyone edits vs reviewed change + version pin
- Model update policy: who owns `ollama pull`, who re-runs eval before deploy
- Rollback path when a new prompt/model regresses answers

Answer:

Part C — Enhancements (scope & prioritize, not blockers)

C1 — Which NEW tools should the agent get? (rank them) ⚡

- Cost/tariff tool (kWh → ₹) — needs tariff data: do we have it?
- Weather/ambient tool (wet-bulb → explains tower approach)
- Maintenance-history tool — needs a CMMS/log source
- Forecast tool (expose the existing forecaster to the agent)
- Production-schedule tool (explains load swings)

Answer (ranked + which data exists):

C2 — What documents should the AI cite, in what formats, and who keeps them correct? ⚡

- Curated markdown playbooks (*current*)
- OEM manuals (PDF) / O&M procedures (Word)
- SharePoint/Confluence wiki → connector + auth

Answer:

C3 — How long should the AI remember, and is cross-session memory worth it? ⚡

- Session-only (*current*)
- Remembers prior findings per equipment across sessions
- Remembers operator preferences (₹ vs kWh, terse vs verbose)

Answer:

C4 — Should the AI be PROACTIVE, and where should output land? ⚡

- Scheduled morning brief at shift start
- Anomaly-triggered plain-English alert
- 3-day trend-watch digest
- Land it in: app / Slack / email?

Answer:

C5 — Is image understanding (vision) a real need or a demo nicety? ⚡

- Read a nameplate photo → fill the catalog
- Read a gauge/HMI panel photo
- Critique a pasted trend screenshot

Answer:

C6 — Do we commit to a / feedback loop now?

- Yes → capture verdicts now so the learning loop + future fine-tune have data
- Failed answers become new eval cases
- Not yet

Answer:

C7 — Which delivery surfaces do operators and management actually want?

- Web app only (*current*)
- Slack/Teams brief + alerts
- Weekly management PDF by email

Answer:

C8 — Is a second facility coming, and when? (*decides build-tenancy-now vs later*)

- Unicharm only for 6+ months → don't build multi-tenant yet
- Another site (e.g. Varanasi Airport) soon → start per-tenant catalog/prompts/KB now

Answer:

C9 — Single agent, or multiple specialized agents collaborating?

- One agent with all tools (*current — modes share one loop*)
- A planner/orchestrator dispatching specialist agents in parallel, then synthesizing (*orchestrator exists — lean on it more?*)
- Specialist sub-agents (efficiency expert, maintenance expert, cost expert)

Answer:

C10 — What compliance, audit-trail retention, and access control is needed?

- Audit log of every Q&A is kept (*exists*) — how long, and who can read it?
- Any framework that gets audited (ISO 27001 / SOC2 / internal)?
- Identity/roles: stays a shared no-auth kiosk (*current*) vs need to know who asked/approved

Answer:

Part D — Infra & On-prem operations (the deployment reality)

D1 — Where does inference run, and must it survive a network/WAN outage?

- Ollama on a LAN/Tailscale GPU box (*current — 100.x host over Tailscale*)
- If Tailscale / WAN drops, must the AI keep working on the local LAN? (true air-gap vs convenience VPN)
- Is the GPU box co-located in the plant, or remote?

Answer:

D2 — What's the model-serving & per-task model strategy? ⚙️

- Ollama now (qwen2.5:14b reasoning · llama3.1:8b tools/SQL · llama3.2 auditor · nomic-embed for RAG) *(current)*
- Switch to vLLM at >10 concurrent users or <5s SLA
- Confirm: is per-task right-sizing acceptable, or one model for everything?

Answer:

D3 — Backups, disaster recovery & data retention for the on-prem stack? ⚙️

- What gets backed up: Postgres app data, pgvector embeddings, audit logs? How often?
- Restore target — how fast must we recover if the box dies?
- MySQL telemetry is read-only source (not ours to back up) — confirm

Answer:

D4 — Who operates the on-prem box, and what's the update/maintenance window? ⚙️

- Ownership: who runs `ollama pull`, applies updates, watches Grafana/Prometheus/Loki?
- Plant runs 24/7 — when can we restart Ollama/backend safely?
- On-call expectation if the AI is down overnight, or "use the dashboard" is fine?

Answer:

D5 — Any integration with plant systems (CMMS / BMS / SCADA), and which direction? ⚙️

- Read-only from BMS/SCADA for richer alarms/events? (crosses an OT security boundary)
- Write a work order into a CMMS (Maximo / SAP PM / UpKeep / spreadsheet)?
- Strictly no connection to plant control systems *(current — read-only, no actuation)*

Answer:

Decisions captured (fill in after the session)

#	Decision	Answer	Owner	→ Task / ADR
A1	Highest-value job			
A5	Autonomy ceiling (A/B/C/D)			
B3	Truthfulness gate (hard/best-effort)			
B5	Open-source firm?			
B6	GPU + concurrency			
B7	Pass bar for "trustable"			
C1	New tools (ranked)			
C8	Second facility timeline			
D1	Offline/air-gap requirement			

#	Decision	Answer	Owner	→ Task / ADR
D5	CMMS/BMS integration target			
—	Next-release top 3			

Pair with [FUTURE_TASKS.md](#) to turn answers into work.